

ORIGINAL ARTICLE OPEN ACCESS

Self-Explanation-Guided Virtual Museum Learning for Knowledge Co-Construction Based on the Constructivist Framework: A Randomised Control Study

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Received: 15 August 2025 | **Revised:** 7 April 2026 | **Accepted:** 11 July 2026

Keywords: collaborative learning | constructivist framework | self-explanation | teaching/learning strategies | virtual museum

ABSTRACT

Background: Museum learning is an important approach to facilitating cross-disciplinary knowledge integration. To expand the accessibility of museum education and to enhance learning efficiency, virtual museums are aiming to overcome the constraints of time, crowds, and the limited resources of physical venues by providing a more flexible and immersive digital learning environment.

Objective: Scholars have suggested using the 4C elements (i.e., context, construction, conversation, collaboration) based on the constructivist framework to facilitate knowledge co-construction. On the other hand, scholars have also pointed out that, in the 4C learning context, students often lack experience and the ability to think deeply, hence leading to disappointing outcomes. Several previous studies have reported the problems faced in team work, such as the free-rider effect among members, which can negatively affect collaborative tendencies and the collective efficacy of the team.

Method: To address these issues, this study proposed a self-explanation-guided virtual museum learning (SE-VML) model, which guides individual learners to link new knowledge with their existing cognitive models, and to construct and modify their own mental models to enhance collaborative effectiveness. To test the effectiveness of this approach, a true experimental design was conducted by randomly assigning 56 sixth-grade students to an experimental group using the SE-VML model and a control group using the conventional virtual museum learning (C-VML) model.

Results and Conclusions: Results showed that, compared with the control group, the experimental group obtained significantly higher post-test scores in learning achievement, critical thinking tendency, collaboration tendency, and metacognition. In addition, behavioural sequence analysis indicated that students in the experimental group showed more frequent rereading behaviours and more goal-oriented exploration. Interview findings further suggested more favourable perceptions of the learning experience among experimental group students.

1 | Introduction

Museum learning is an exploration experience in which learners gain background knowledge of artefacts and acquire rich knowledge of history, science, art, language, and other

interdisciplinary subjects (Lin et al. 2023; Zhou et al. 2022). In the conventional approach, the implementation of museum-based learning faces several limitations, including the need for advance reservations, the issue of overcrowding that hinders the viewing experience, and the significant time and travel costs

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Practitioner Notes

What is Already Known About This Topic

- Virtual museums can overcome the temporal and spatial constraints of physical museums, providing a more flexible and immersive learning environment.
- Scholars have indicated the importance of using the constructivist framework (e.g., the 4C elements) to guide knowledge construction.
- In collaborative learning, the ‘free-rider effect’ often occurs due to a lack of individual deep thinking, negatively affecting the effectiveness of teamwork.

What This Paper Adds

- A self-explanation-guided virtual museum learning (SE-VML) model is proposed, which incorporates self-explanation as a preparatory activity before collaborative tasks.
- Based on the constructivist framework, this model uses structured guiding questions to prompt students’ self-explanation to deepen their individual understanding.
- A randomized control study verified that the SE-VML model, compared to the conventional approach, more effectively improves students’ learning achievement, critical thinking tendency, collaboration tendency, and metacognition.

Implications for Practice and/or Policy

- Incorporating guided self-explanation activities before group collaboration can effectively enhance students’ learning achievement, critical thinking tendency, collaboration tendency, and metacognition.
- This approach helps ensure that students are better prepared for teamwork, thus promoting a more effective and equitable collaborative process.
- It is recommended that this approach be applied to other courses and learning contexts.

involved (Islek and Danju 2019; Lee et al. 2020; Ratumbuisang et al. 2024; Tatlı et al. 2023). Virtual museums can overcome these limitations to some extent and offer rich content. However, without effective support to facilitate learners’ cognitive engagement and reflective experiences, this content is often difficult to transform into meaningful learning (Hu and Hwang 2024). Thus, scholars have recommended implementing virtual museum learning based on the constructivist framework to engage learners in the contexts with the 4C (Context, Construction, Conversation, Collaboration) elements: providing real-world contexts, enabling exploration and practice to facilitate knowledge construction, facilitating multiple perspectives through conversation, and encouraging sharing and collaboration (Feyzi Behnagh and Yasrebi 2020; Murphy 1997; Rajaonarivo et al. 2019). In particular, employing concept mapping to facilitate knowledge co-construction in virtual learning contexts has been recommended by several previous studies (Jia et al. 2024; Kim and Ju 2023).

However, researchers have also pointed out the need to be aware of the challenges learners often face while collaborating under the constructivist framework. For example, learners are often unable to bring their own initial perspectives into group conversations due to a lack of independent thinking (Baloche and Brody 2017; Puntambekar 2006; Tu 2025). Learners may also have difficulty effectively expressing their understanding due to differences in knowledge structure and comprehension levels, resulting in the collaboration not proceeding smoothly (Lawson and Mayer 2021). In this case, the free-rider effect may also occur, where members become less committed to the learning task and rely on their peers to complete it, reducing collaboration tendencies and affecting overall learning outcomes (Liang et al. 2021). These collaborative challenges become more salient in information-rich virtual museum settings (De Back et al. 2023; Zhou et al. 2022). Under a constructivist framework, the focus of virtual museum learning shifts from simple information acquisition to a complex task of collaborative multimodal meaning-making (Ho et al. 2011; Wehrmann and Zender 2025). In this process, personal understanding plays a vital role in effective collaboration. When learners lack a well-organised personal understanding of the learning content, the collaborative challenges described above are further intensified (Hsu et al. 2016; Petersen et al. 2023). Therefore, before engaging in dialogue and collaboration, it is necessary to supplement learning with individual scaffolding strategies to enhance learners’ personal understanding and ensure the effectiveness of subsequent group discussions (Hsu et al. 2016; Puntambekar 2006; Roy and Chi 2005).

Self-explanation, as a generative learning strategy, can help learners select, organise, and structure learning content to enhance their understanding of key knowledge (Fiorella and Mayer 2016). For example, Wang et al. (2023) showed that the self-explanation strategy is effective in terms of activating deep generative cognitive processing, which helps learners acquire knowledge. Moreover, especially in the information-rich contexts of virtual reality (VR), learners may become overly concerned with the sensory richness of the environment, resulting in their limited cognitive information processing capacity being occupied (Mayer et al. 2022). In this context, self-explanation activities can be effective in terms of guiding learners to focus on critical learning content, thus enhancing the effectiveness of learning (Makransky 2022).

The core of self-explanation is to motivate learners to actively integrate new information with existing knowledge, and to fill knowledge gaps by targeting what is not understood (Roy and Chi 2005; Ting et al. 2018). When learners try to explain a concept, they are able to identify whether their understanding of the knowledge is complete, and can enhance their learning achievement by revising and supplementing their mental models (Pi et al. 2022). In this process, metacognition plays a key role in helping learners to observe and adjust their own thinking processes, that is, to “think about thinking” (Schwartz et al. 2016). Moreover, to build a complete explanation, learners must analyse information, examine relationships among concepts, and organise a logical argument. This process strengthens critical thinking (Fiorella 2023; Hand et al. 2021). Most importantly, when learners develop personal understanding through self-explanation, they enter collaborative settings with greater confidence and capacity to articulate their views, thereby

enhancing their collaboration tendencies (Hsu et al. 2016; Ting et al. 2018).

In virtual museum learning based on the 4C framework, learners are required to interpret, evaluate, and integrate multimodal information (e.g., text and images) to construct shared understanding (Ho et al. 2011; Hu and Hwang 2024). To participate effectively in this process, learners need not only a collaborative disposition, but also critical thinking to evaluate and synthesise ideas, and metacognitive strategies to monitor and regulate their learning (Chiu et al. 2023; Hu and Hwang 2024). Therefore, rather than directly examining interaction quality, this study focused on learner-level cognitive and dispositional factors that underpin effective collaboration, including personal understanding, critical thinking tendency, collaboration tendency, and metacognition. Based on this perspective, this study proposed a self-explanation-guided virtual museum learning (SE-VML) model. To verify the effectiveness of this model, the following research questions were explored:

1. Is there a significant difference in learning achievement between the SE-VML model and the conventional virtual museum learning (C-VML) model?
2. Is there a significant difference in critical thinking tendency between the SE-VML model and the C-VML model?
3. Is there a significant difference in collaboration tendency between the SE-VML model and the C-VML model?
4. Is there a significant difference in metacognition between the SE-VML model and the C-VML model?
5. What are the differences in learning behaviour patterns and perceived experiences between the two groups of students using the different learning models?

2 | Literature Review

2.1 | Virtual Museum

The sense of presence and immersion created by virtual museums can effectively overcome the temporal and spatial constraints of traditional museum learning (Islek and Danju 2019; Lee et al. 2020) and provide learners with a personalised learning experience (Lin et al. 2023; Zhou et al. 2022). Despite the unprecedented interactivity and sensory experience that virtual museums bring to learners, the core challenge remains how to effectively motivate and guide students to actively construct knowledge (Shehade and Stylianou-Lambert 2020). To date, most studies have focused on the technical development of virtual museums, emphasising the enhancement of the immersive experience of users, but paying less attention to their actual needs and learning effectiveness (Tatlı et al. 2023; Wu et al. 2023).

The lack of clear learning objectives and structured guidance in many virtual museums may cause learners to get lost in the richness of information, overload their cognitive load, and ultimately become passive receivers of information or even lose their motivation to learn (Hu and Hwang 2024; Karakuş Yılmaz et al. 2025). This highlights the fact that a highly immersive

experience is not the same as efficient learning, and without strategic guidance, it is difficult to transform rich content into meaningful knowledge.

Scholars have pointed out that the key to successful virtual museum learning is to guide learners in organising and comprehending information (Kayaalp et al. 2024; Lin et al. 2023; Puig et al. 2020). For example, Chiu et al. (2023) used experiential learning theory as a framework to help learners enhance their art appreciation skills, and Hu and Hwang (2024) proposed the self-adapted mobile concept mapping-based problem-posing (CMPP) method to help students change from passive information receivers into active inquirers. These studies confirm the feasibility of integrating learning strategies into virtual museum learning.

However, scholars have further emphasised that meaning making involves learners interpreting and connecting information within the socio-cultural context of the exhibition (Kayaalp et al. 2024; Puig et al. 2020). It also involves actively processing and organising acquired information and constructing their own mental models (Fiorella et al. 2020; Wu et al. 2023). This process requires a strategy that can directly trigger learners' internal cognitive processes. Such a strategy should guide them to reason about the information, fill logical gaps, clarify causal relationships, and internalise new knowledge into their existing cognitive structures (Makransky 2022; Wen et al. 2024). Petersen et al. (2023) and Pi et al. (2022) found that in VR learning, self-explanation facilitates this inside-out transformation process, organising fragmented information into coherent personal mental representations. It is a vital learning strategy for helping learners focus on, organise, and internalise critical content (Elme et al. 2022; Fiorella et al. 2020).

Therefore, integrating self-explanation into virtual museum learning is a key way to foster meaningful learning. By combining pedagogical strategies with a structured, interactive environment that motivates learners to explore, reflect, and organise information, the potential of virtual museums can be truly transformed into a purposeful and profound learning experience (Chiu et al. 2023; Karakuş Yılmaz et al. 2025; Seprum and Wongwatkit 2025).

2.2 | Self-Explanation

Self-Explanation Theory (Chi et al. 1989) refers to the idea that learning is enhanced when learners generate explanations while studying new information or solving problems. Following this theory, self-explanation has been recognised as a potential learning strategy emphasising the need to guide learners to actively construct knowledge by linking existing cognitive schema with new knowledge to facilitate meaningful learning (Bisra et al. 2018; Fiorella et al. 2020; Schwartz et al. 2016). The process involves learners focusing on key information, organising it into coherent mental structures within limited working memory, and integrating it with long-term memory (Fiorella et al. 2020; Sweller 2011).

Self-explanation can be presented in a variety of forms, such as oral, written, or graphical, and even if the content is not

entirely complete or correct, the process itself helps learners extract key information and generate new knowledge (Bisra et al. 2018; Zhang et al. 2024). Studies have found that oral explanations (Jacob et al. 2020), written explanations (Lawson and Mayer 2021), drawing aids (Zhang et al. 2024), and scaffold-guided question answering (Wen et al. 2024) can significantly enhance learning achievement. Among these methods, logically structured guided questions have been shown to be effective in terms of triggering deep cognitive processes of self-explanation (Buchner 2021; Wen et al. 2024), while open-ended prompts that involve higher cognitive processing have also been found to be effective (Bisra et al. 2018).

In collaborative digital learning contexts, self-explanation has been validated as an effective strategy that enhances learners' metacognition, critical thinking, and collaboration tendency (Hand et al. 2021; Hsu et al. 2016; Ting et al. 2018). The mechanism of self-explanation is closely related to the construction of a mental model, whereby learners construct and revise their internal cognitive models by focusing on, selecting, organising, and integrating information (Elme et al. 2022; Pi et al. 2022; Schwartz et al. 2016). Therefore, this process not only facilitates metacognition, allowing learners to recognise gaps in their understanding, but also effectively enhances the ability to apply concepts through both near transfer and far transfer, thereby facilitating the integration of new knowledge (Fiorella and Mayer 2016; Lawson and Mayer 2021). Moreover, in collaborative learning contexts, self-explanation prompts individuals to externalise their internal understanding as explicit knowledge representations, establishing a shared basis for meaning negotiation (Hand et al. 2021; Hsu et al. 2016; Ting et al. 2018). Peer comparison and reciprocal feedback stimulate reflection and repair of existing representations and, in turn, strengthen critical thinking by fostering evaluation of the value and plausibility of information (Fiorella 2023; van Peppen et al. 2018).

Besides, self-explanation activities are particularly important in sensory-rich learning contexts such as VR, as they help learners resist environmental interference and focus their cognitive resources on core learning content to reduce cognitive load and enhance learning effectiveness (Elme et al. 2022; Makransky 2022). However, prior research on VR has seldom incorporated self-explanation strategies, and their application remains limited (Elme et al. 2022; Petersen et al. 2023). Petersen et al. (2023) also pointed out that in VR, merely asking individual students to perform silent, imagined summarisation (a form of covert self-explanation) did not result in significantly different learning outcomes compared to a control group with no such activities. This finding revealed that in VR, it is insufficient to merely expect learners to spontaneously engage in deep cognitive processing.

Therefore, this study used students' written self-explanations as the primary learning intervention in the virtual museum learning. It guided learners to understand the learning content, actively select key messages, and organise the content. It also helped them integrate the content with their prior knowledge. This process facilitated deep learning and knowledge transfer.

3 | Self-Explanation-Guided Virtual Museum Learning Model

Based on the four core elements of Murphy's (1997) constructivist framework of context, construction, conversation, and collaboration, and with reference to Buchner (2021) and Elme et al. (2022), a self-explanation-guided virtual museum learning model was designed. In the proposed model, self-explanation does not merely function as a generative learning strategy. Rather, it operates as a cognitive regulation mechanism that constrains exploration, stabilises attention, and supports the transformation of multimodal sensory input into conceptual representations.

Specifically, by requiring learners to articulate their understanding in written form, self-explanation regulates the otherwise open-ended exploratory nature of VR by imposing cognitive checkpoints that prompt the selection of relevant information, the organisation of ideas, and coherence checking. This process reduces attentional diffusion caused by sensory richness and helps learners align exploration with learning goals.

In the VR scenario, learners were guided to engage in self-explanation based on systematic guiding questions to facilitate reflection and knowledge organisation, thereby deepening their understanding and construction of the learning content. Across the four stages of the model, self-explanation serves as a bridge between individual sense-making and subsequent social knowledge co-construction, ensuring that learners enter collaborative phases with cognitively prepared and explicit representations. The model consists of four stages, as shown in Figure 1. Each stage operationalises a distinct regulatory function, progressively guiding learners from unstructured exploration towards coordinated collaborative knowledge construction. The following sections explicitly highlight the logical progression between stages and how cognitive readiness established in earlier stages lays the foundation for subsequent phases.

In the context stage, this study utilised open data such as artefacts and images provided by the National Palace Museum in

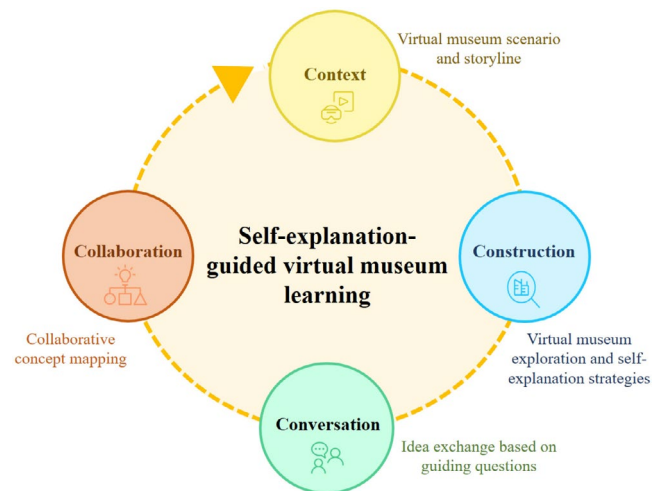


FIGURE 1 | Self-explanation-guided virtual museum learning model.

Taipei to construct a virtual museum as the learning environment. This environment helped learners study actual artefacts in the museum while overcoming time, space, and geographic constraints. The rich contextualisation of the virtual museum provided learners with a contextualised learning experience. Through repeated exploration of the knowledge of the artefacts and the interactive feedback mechanism of the scenarios, learners were able to not only view the artefacts but also gain a deeper understanding of their historical, artistic, and cultural contexts, thereby facilitating cross-disciplinary knowledge integration. Overall, the learning environment of the virtual museum is realistic, highly interactive, and educational (see Figure 2). This stage established learners' initial familiarity with and exploratory experience of the artefacts, providing a concrete sensory and contextual foundation for the subsequent construction stage.

Construction stage: A learning activity that involved exploration and self-explanation of the virtual museum. First, learners

actively explored the background knowledge of the artefacts through structured guiding questions embedded in each scene of the virtual museum. The system provided real-time interactive feedback to assist learners in initially checking their understanding. These guiding questions also prompted learners to think about how new knowledge was organised and how it linked to their previous experiences. Through feedback from the system, learners were able to review the correctness of the knowledge content and actively construct knowledge instead of passively accepting information (Figure 3). Notably, the structured guiding questions provided real-time feedback so that learners could secure an accurate initial grasp of the artefacts' factual information. Afterwards, the self-explanation task asked learners to digest, organise, and internalise these verified ideas. Learners constructed coherent meaning in their own words. By design, this task offered no instant system feedback; its aim was to foster an individual generative cognitive process.

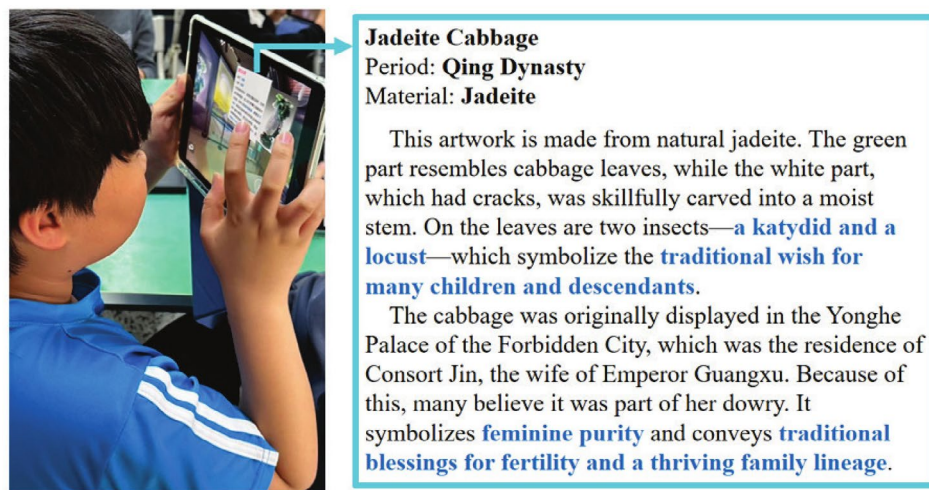


FIGURE 2 | Contextual learning in the virtual museum.

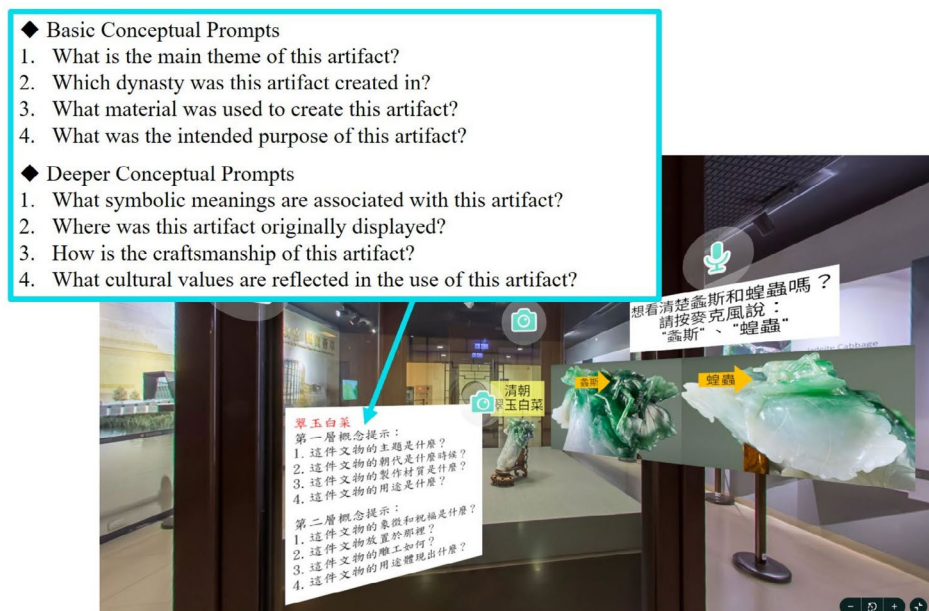


FIGURE 3 | Structured guiding questions.

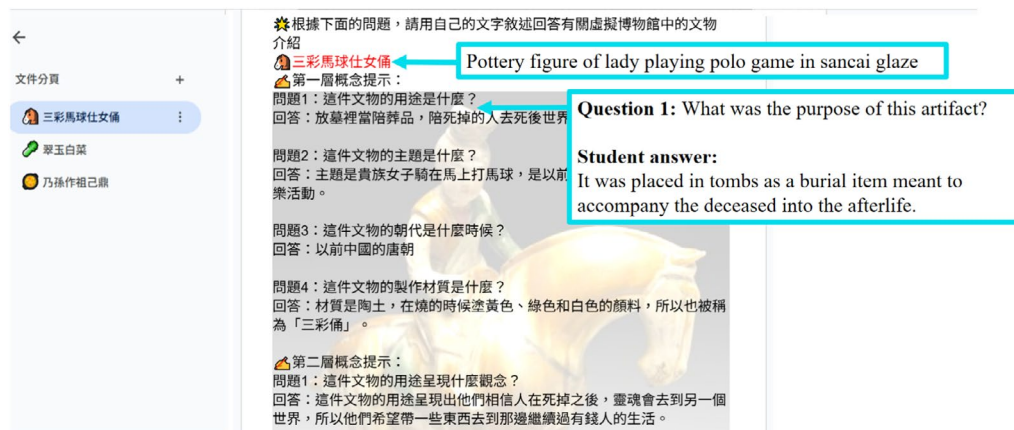


FIGURE 4 | Self-explanation task.

Next, in order to deepen knowledge construction, this study adopted the self-explanation-guided virtual museum learning model, which allowed learners to complete self-explanation tasks in textual form through online structured guiding worksheets (Figure 4). The core of the self-explanation strategy was to motivate learners to externalise their own understanding of knowledge, which not only reduced the burden on their working memory but also enhanced their participation in the construction of meaning in the content, generating clear and detailed descriptions of knowledge for themselves. Specifically, this process required learners to organise and reorganise content, making meaningful connections between new information and existing knowledge. In this study, “connecting existing information with new concepts” was operationalised through artefact appreciation. Students already possessed fragmented understandings from class and supplements, for instance, knowledge of dynasty context and social customs (e.g., the Tang dynasty’s emphasis on burial items), the symbolism of idioms (e.g., 螽斯衍慶, denoting many descendants), and the evolution of scripts and inscriptions (e.g., text cast on the inner walls of bronze vessels). In virtual museum learning, learners were situated in an artefact appreciation context. They activated and reorganised their prior knowledge and then integrated it with new, artefact-specific information via self-explanation tasks. The process of translating understanding into words facilitated learners’ knowledge construction, and enhanced the retention and subsequent transfer of knowledge.

The key output of this construction stage was the written record produced by learners through the self-explanation task. These externalised personal understandings served not only as concrete outcomes of individual knowledge construction but also as the cognitive readiness that supported the subsequent conversation and collaboration stages. In other words, the outputs of self-explanation provided shareable and discussable materials for later social interaction.

Conversation stage: Learners shared their ideas, discussing and reflecting on the content they had completed in the self-explanation task with their group members. During the process, each learner was required to share their personal insights with the group and to examine their depth of understanding of

the content through oral elaboration and clarification via mutual questioning. Social interaction was used to promote the refinement of personal knowledge, allowing learners to negotiate meaning and develop a more complete and detailed body of knowledge through the exchange and collision of ideas. Notably, the conversation stage did not start from scratch. It was grounded in the written outputs produced during the self-explanation task in the construction stage. Learners entered group discussions with their already organised personal understanding. Therefore, the outputs of self-explanation served as a cognitive bridge, transforming individual internal construction into a shared starting point for social interaction.

Collaboration stage: Group members constructed a concept map of knowledge through synchronous collaboration, shared the results of their group discussions (Figure 5), presented their work to the class, and received feedback and suggestions from both peers and the teacher. The collaborative process not only required members to organise and connect the core concepts together but also facilitated the externalisation and sharing of knowledge through visual images. Through collaborative planning and practice, group members achieved collaborative knowledge construction and deepened their systematic understanding of the content. Collaborative knowledge construction at this stage benefited from the refined understanding and clarification achieved during the conversation stage. Through the scaffolding provided by the preceding construction and conversation stages, learners entered the collaboration phase with sufficient cognitive readiness and a shared foundation of understanding. This enabled the final collaborative concept mapping to proceed efficiently.

In summary, the four stages of this model presented a cognitive progression pathway. The context stage provided an exploratory foundation. The construction stage produced externalised personal understandings through self-explanation. The conversation stage transformed these personal understandings into shared social knowledge. The collaboration stage systematically synthesised the shared understanding into visualised concept maps. The outputs of self-explanation served as the critical hinge connecting these four stages. This ensured that cognitive readiness was successfully transferred from the individual level to the social interaction level.

Figure 6 shows the system architecture, which consists of a teacher management platform, a database, and a student learning platform.

The teacher management platform includes two modules. The virtual museum teaching module provides scene resources,

structured-prompted self-explanation, and collaborative concept mapping resource management. This empowers the teacher to design instruction and tailor task difficulty based on students' needs. The virtual museum management module manages student accounts and collaboration privileges, enabling the teacher to monitor account usage and ensure smooth learning activities.



FIGURE 5 | Collaborative concept mapping.

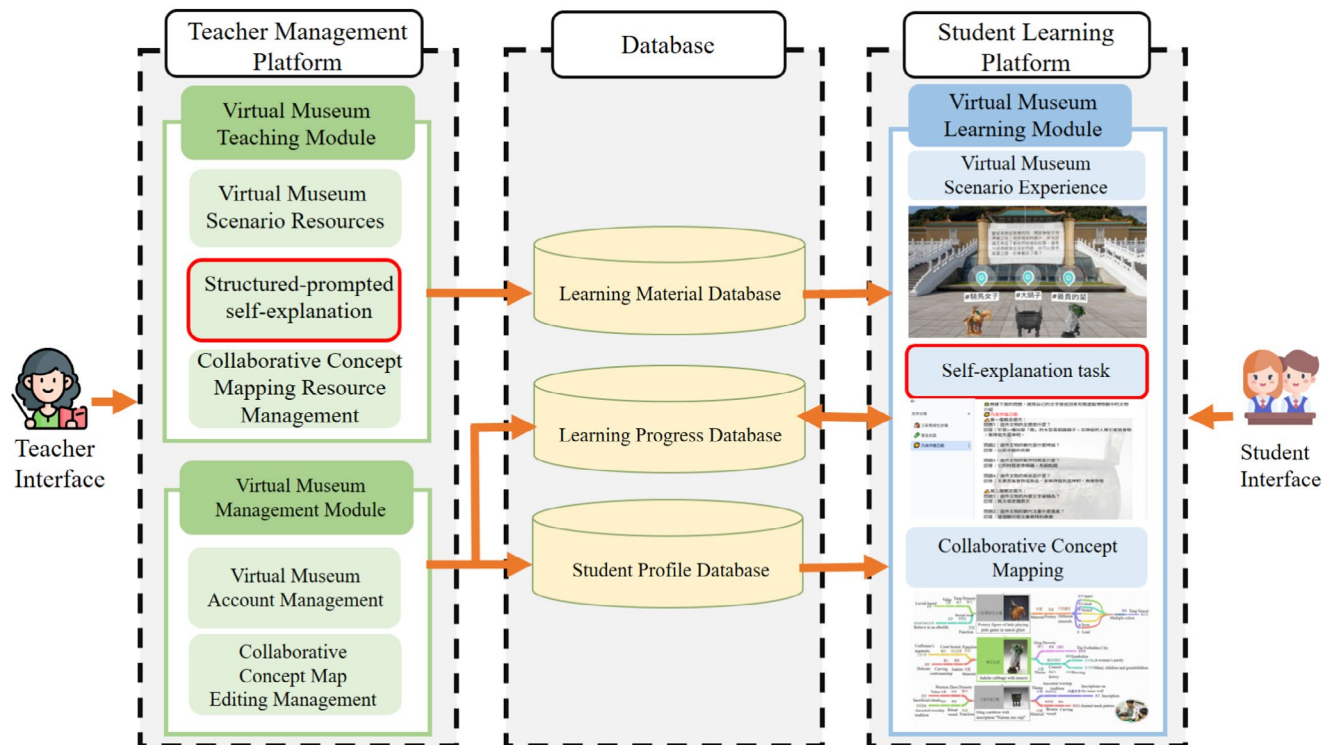


FIGURE 6 | Structure of the SE-VML system.

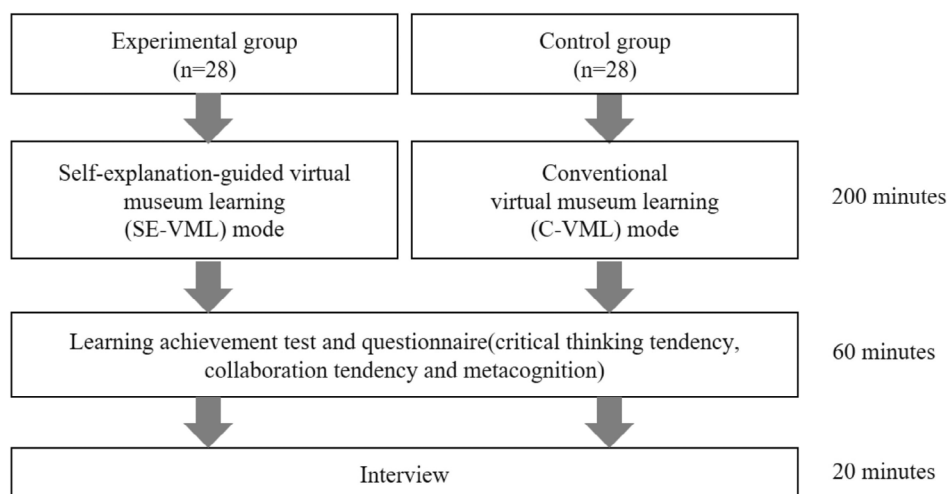


FIGURE 7 | Experimental procedure.

The database has three components. The learning material database stores teaching content, serving as a central repository to ensure students access consistent information. The learning progress database records student interactions and progress. This enables the teacher to monitor learning status and provides data for subsequent feedback and analytics. The student profile database manages student accounts and passwords, which ensures system security and supports personalised learning tracking.

The student learning platform is a virtual museum learning module that allows students to experience virtual museum scenarios, perform self-explanation tasks, and complete collaborative concept mapping. It supports a complete learning cycle from exploration and self-explanation to collaborative concept mapping.

Learners logged into the student learning platform and entered the virtual museum learning module to begin the virtual museum scenario experience. The learning progress database recorded their exploration paths and interaction behaviours. During exploration, the structured guided questions were triggered. After submitting responses, learners received immediate interactive feedback.

Learners then proceeded to the self-explanation task. They articulated their understanding in writing to externalise their internal thinking and form coherent knowledge representations. All content was stored in the learning progress database. Next, learners shared self-explanation outputs for group discussion. Finally, groups used collaborative concept mapping with synchronous collaboration tools to build a knowledge concept map. They presented the map to the class and received peer and instructor feedback.

In contrast, the conventional virtual museum learning (C-VML) model did not require learners to externalise or organise their understanding individually prior to collaboration, leaving cognitive preparation for group discussion largely implicit.

4 | Experimental Design

4.1 | Participants

Participants were drawn from sixth-grade students at an elementary school in southern Taiwan. Using a simple random assignment procedure (without stratification), students were allocated to either the experimental condition (SE-VML; $n = 28$) or the control condition (C-VML; $n = 28$).

Both groups used the same learning content and received structured guided questions as scaffolds. The key difference between the two conditions lay in the learning process: students in the experimental group were required to complete an explicit self-explanation task to externalise and organise their understanding before engaging in group collaboration, whereas students in the control group proceeded directly to collaborative concept mapping without this individual knowledge construction step.

4.2 | Experimental Procedure

This study adopted a randomised controlled (true experimental) design, in which random assignment was used to minimise pre-existing differences between groups. Therefore, there was no pre-test in the procedure. While randomisation does not guarantee perfect equivalence in small samples, it is widely regarded as an appropriate method for establishing baseline comparability in experimental research (Keppel and Wickens 2004).

Figure 7 shows the experimental process. The experimental group and the control group each took part in a 200-min learning activity, which included the teacher's explanation of the operation of the virtual museum equipment, the function of the collaborative conceptual mapping tool, and the experience of the overall activity flow. Students in the experimental group learned with the SE-VML model, and students in the control group learned with the C-VML model. Students

were then given 60 min to complete the learning achievement test, as well as the critical thinking tendency questionnaire, collaboration tendency questionnaire, and metacognition questionnaire. Finally, a 20-min interview was conducted to further investigate participants' perceptions of the proposed methodology.

4.3 | Measuring Tools

The learning achievement test was designed by two elementary school teachers with more than 5 years of teaching experience. The test focused on assessing learners' background knowledge of artefacts, covering cross-disciplinary content from the language arts and social sciences domains, and consisted of 10 multiple-choice questions with a total score of 100 points. The KR-20 reliability coefficient was 0.85.

The critical thinking tendency questionnaire was administered using a scale by Lin et al. (2019), adapted from the Chai et al. (2015) scale, and included six items, such as, "During the learning process, I will assess the reasonableness of different opinions." Using a 5-point Likert scale, the Cronbach's alpha of the questionnaire was 0.86.

The collaboration tendency questionnaire was based on the Lai and Hwang (2014) scale and included six items, for example, "When collaborating with my peers, I usually have good communication with them." The Cronbach's alpha of the questionnaire was 0.91, using a 5-point Likert scale.

The metacognition questionnaire, adapted from Lai and Hwang's (2014) scale, consisted of five items, such as, "I regularly check to see whether or not I have achieved my goals." Using a 5-point Likert scale, the Cronbach's alpha of the questionnaire was 0.82.

For the analysis of learning behaviour patterns, the learning behaviour coding scale proposed by Hwang et al. (2023) was used as a reference and was adjusted to fit the context of this study. Based on the design of the virtual museum learning environment, the following eight behavioural codes were selected and retained: start experience (SE), explore object (EO), repeat object (RO), answer correctly (AC), wrong answer (WA), read materials (RM), next scene (NS), and finish experience (FE). The definitions and descriptions of these behaviours are shown in Table 1. These codes were mainly used to classify the behaviours displayed by students during the process of completing the learning tasks.

For the perceptual learning experience, the interview outline was adapted from Hwang et al. (2009), and included questions such as: "What is the difference between using this way of learning and the normal way of learning? How do you think it helps your learning?" Face-to-face semi-structured focus group interviews were conducted with 10 students from the experimental group and 10 from the control group. The process of analysing the interview results was conducted with the support of ATLAS.ti.22. As suggested by Creswell and Poth (2016), open coding was used. Similar items in the code

TABLE 1 | Coding table for learning behaviour.

Code	Definition	Description of behaviour in the VML
SE	Start Experience	Start a new experiential activity
EO	Explore Object	Click on the description or a guiding object
RO	Repeat Object	Repeatedly click on the description or a guiding object
AC	Answer the question correctly	Correctly answer the guiding question
WA	Wrong Answer	Select a wrong answer
RM	Read Materials	Read or listen to learning materials
NS	Next Scene	Transfer from one scene to the next one
FE	Finish Experience	Complete the entire learning experience

TABLE 2 | The independent samples *t*-test result of learning achievement.

Groups	<i>N</i>	Mean	SD	<i>t</i>	<i>d</i>
Experimental	28	82.86	11.17	2.30*	0.61
Control	28	75.36	13.19		

* $p < 0.05$.

list were grouped into clusters. These clusters formed new categories, which were further generalised into themes and subthemes. This approach provided a structured framework for analysis.

5 | Results

5.1 | Learning Achievement

In this study, the independent samples *t*-test was conducted on students' learning achievement to examine the differences between the two groups of students after learning with the different learning modes. The normality of the data was tested using the Shapiro–Wilk method, and the results showed that these data conformed to the normal distribution ($p > 0.05$).

As shown in Table 2, the mean and standard deviation of the experimental group were 82.86 and 11.17, respectively, while those of the control group were 75.36 and 13.19, respectively. The experimental group's learning achievement was significantly better than that of the control group ($t = 2.30$, $p = 0.03 < 0.05$), with a medium effect size (Cohen's $d = 0.61$) (Cohen 1988).

5.2 | Critical Thinking Tendency

The independent samples *t*-test was used to analyse the effect of learning mode on students' critical thinking tendency. The normality of the data was tested using the Shapiro–Wilk method, and the results showed that these data conformed to the normal distribution ($p > 0.05$).

As shown in Table 3, the mean and standard deviation of the experimental group were 4.37 and 0.54, respectively, while those of the control group were 4.03 and 0.62, respectively. The experimental group's critical thinking tendency was significantly higher than that of the control group ($t = 2.22$, $p = 0.03 < 0.05$), with a medium effect size (Cohen's $d = 0.58$) (Cohen 1988). It is inferred that the SE-VML model was effective in terms of helping students enhance their critical thinking tendency.

5.3 | Collaboration Tendency

The independent samples *t*-test was used to analyse the effect of learning mode on students' collaborative tendency. The normality of the data was tested using the Shapiro–Wilk method, and the results showed that these data conformed to the normal distribution ($p > 0.05$).

As shown in Table 4, the mean and standard deviation of the experimental group were 4.49 and 0.69, respectively, while those of the control group were 4.12 and 0.63, respectively. The collaborative tendency of the experimental group was significantly higher than that of the control group ($t = 2.11$, $p = 0.03 < 0.05$), with a medium effect size (Cohen's $d = 0.56$) (Cohen 1988). It is inferred that the SE-VML model was effective in terms of helping students enhance their propensity to collaborate.

5.4 | Metacognition

The independent samples *t*-test was used to analyse the effect of learning mode on students' metacognition. The normality of the data was tested using the Shapiro–Wilk method, and the results showed that these data conformed to the normal distribution ($p > 0.05$).

As shown in Table 5, the mean and standard deviation of the experimental group were 4.58 and 0.59, respectively, while those of the control group were 4.18 and 0.66, respectively. The experimental group's metacognition was significantly higher than that of the control group ($t = 2.40$, $p = 0.02 < 0.05$),

TABLE 3 | The independent samples *t*-test result of critical thinking tendency.

Groups	<i>N</i>	Mean	SD	<i>t</i>	<i>d</i>
Experimental	28	4.37	0.54	2.22*	0.58
Control	28	4.03	0.62		

* $p < 0.05$.

with a medium effect size (Cohen's $d = 0.64$) (Cohen 1988). It is inferred that the SE-VML model proposed in this study was effective in terms of helping students enhance their metacognition.

5.5 | Behavioural Pattern Analysis

The behavioural codes in this study were developed with reference to the coding table proposed by Hwang et al. (2023) and were adjusted according to the characteristics of the virtual museum developed in this study. The coding scheme included a total of eight behavioural codes, the definitions and descriptions of which are detailed in Table 1. The learning behaviours of the learners were automatically recorded in the virtual museum system and were coded by two researchers according to the coding table, with a Kappa value of 0.80.

The students' behaviours in the virtual museum were analysed using the sequence analysis method to investigate whether the transfer relationship between the experimental group and the control group for each learning behaviour reached statistical significance. In the residual table of learning behaviours, the items listed on the leftmost side represent the starting behaviours, and the items in the upper row represent the subsequent behaviours. *Z*-values were calculated for each possible sequence of behaviours; when the *Z*-value is greater than 1.96, the sequence of behaviours reached the level of significance.

The results of the analysis showed that in the residual table, there were 10 learning behaviours in the experimental group (Table 6) and seven in the control group (Table 7) that reached a significant level. There were seven identical learning behaviours that reached a significant level in both groups and three different significant behaviours.

The behavioural conversion diagram for the experimental group was plotted according to Table 6, as shown in Figure 8, and the behavioural conversion diagram for the control group was plotted according to Table 7, as shown in Figure 9. The values in the graphs represent the *Z*-values of each sequence. The black line represents the significant sequences occurring in

TABLE 4 | The independent samples *t*-test result of collaboration tendency.

Groups	<i>N</i>	Mean	SD	<i>t</i>	<i>d</i>
Experimental	28	4.49	0.69	2.11*	0.56
Control	28	4.12	0.63		

* $p < 0.05$.

TABLE 5 | The independent samples *t*-test result of metacognition.

Groups	<i>N</i>	Mean	SD	<i>t</i>	<i>d</i>
Experimental	28	4.58	0.59	2.40*	0.64
Control	28	4.18	0.66		

* $p < 0.05$.

TABLE 6 | Residual table of the experimental group's learning behavioural analysis.

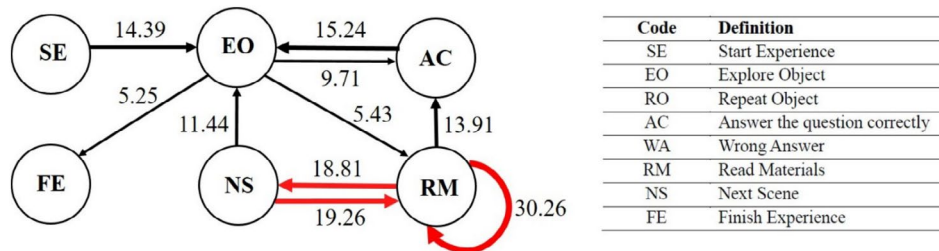
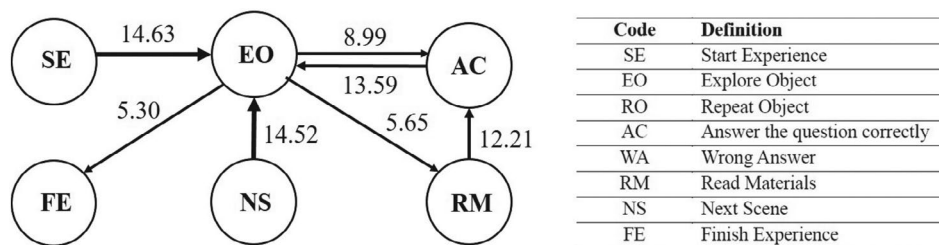
Z-score	EO	RO	AC	WA	RM	NS	FE
SE	14.39*	-4.08	-3.46	-2.54	-2.42	-2.29	-0.55
EO	-5.97	-6.24	9.71*	-0.38	5.43*	-1.55	5.25*
RO	-2.98	-8.11	-0.66	-1.1	-7.52	-4.28	-2.29
AC	15.24*	-4.08	-3.81	-10.21	-2.13	-3.88	-3.73
WA	-0.53	-2.32	-1.25	-0.97	1.75	-2.41	-1.29
RM	-2.24	-6.91	13.91*	-2.67	30.26*	18.81*	-2.27
NS	11.44*	-11.95	-7.88	-8.99	19.26*	-9.71	-3.31

*Z > 1.96.

TABLE 7 | Residual table of the control group's learning behavioural analysis.

Z-score	EO	RO	AC	WA	RM	NS	FE
SE	14.63*	-3.64	-2.97	-2.59	-2.7	-2.34	-0.77
EO	-6.23	-5.91	8.99*	-0.25	5.65*	-1.47	5.3*
RO	-3.12	-7.29	-1.82	-1.23	-6.82	-4.83	-2.42
AC	13.59*	-5.03	-3.73	-9.06	-3.01	-3.79	-3.81
WA	-0.95	-1.92	-1.31	-1.01	1.51	-2.37	-1.34
RM	-3.05	-5.54	12.21*	-2.41	1.67	-0.95	-2.33
NS	14.52*	-11.83	-7.75	-7.57	1.73	-9.69	-2.99

*Z > 1.96.

**FIGURE 8** | The behavioural transition diagram of the experimental group.**FIGURE 9** | The behavioural transition diagram of the control group.

both the experimental group and the control group, while the red line represents the significant sequences occurring only in the experimental group. The arrows indicate the direction of behavioural transition for each sequence, and the thickness of the arrow line reflects the significance and intensity of the behavioural transition.

From Figures 8 and 9, we can observe the same salient behaviours in both groups, including the behavioural paths of Start Experience (SE) → Explore Object (EO) → Read Materials (RM) → Answer the question correctly (AC) → Explore Object (EO), Next Scene (NS) → Explore Object (EO) → Answer the question correctly (AC), and Explore Object (EO) → Finish

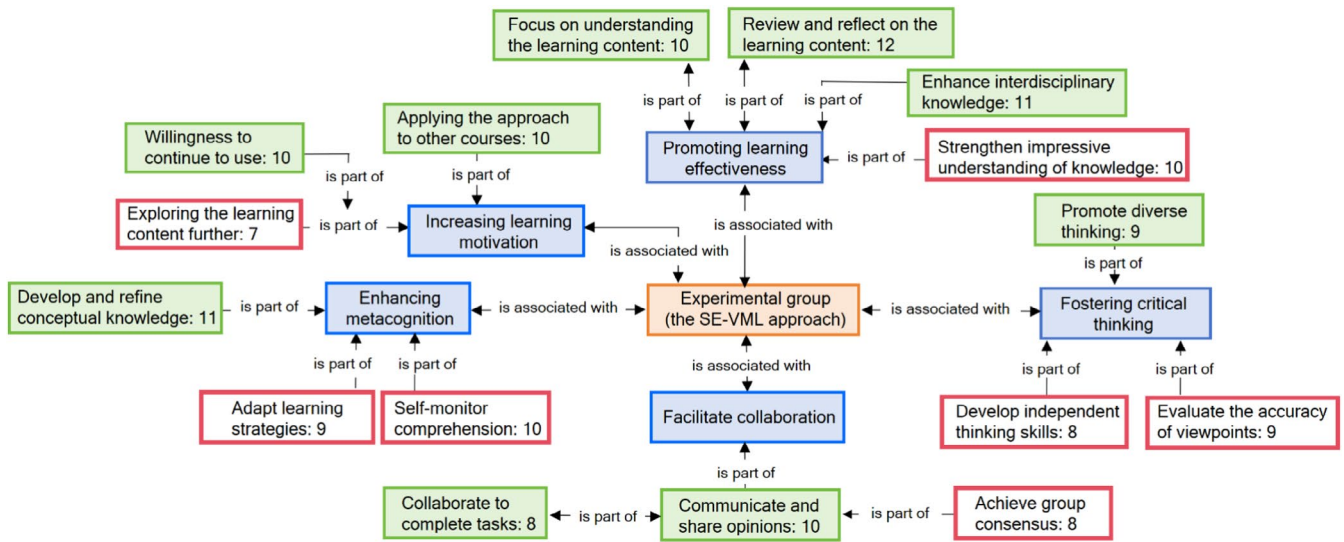


FIGURE 10 | The interview results of the experimental group.

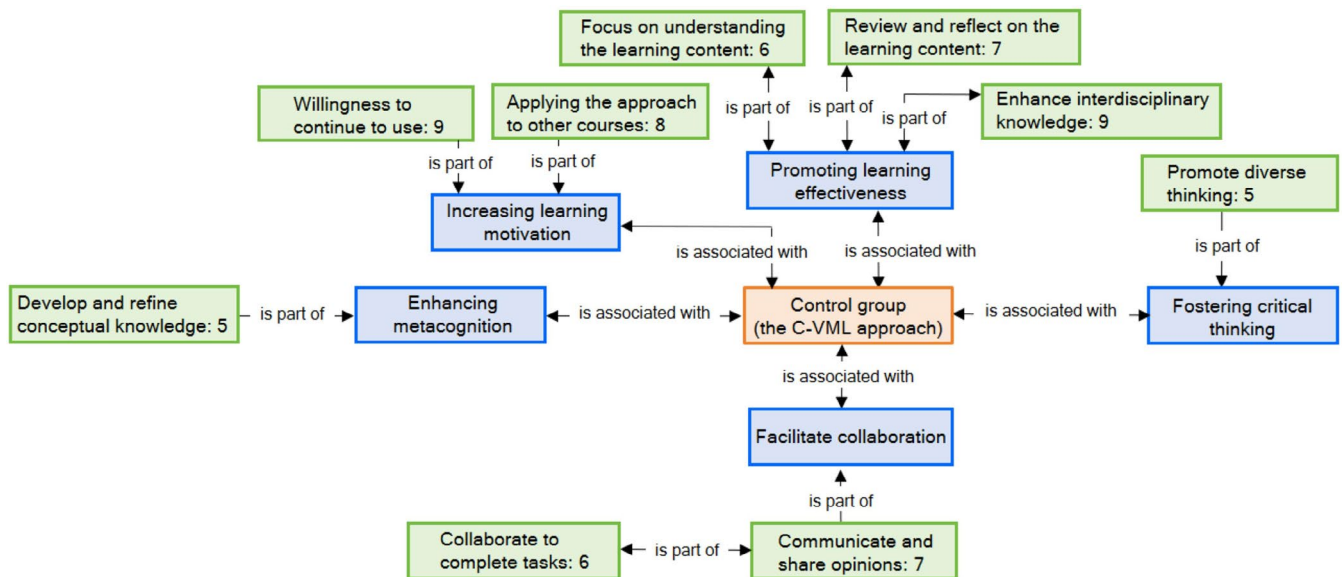


FIGURE 11 | The interview results of the control group.

Experience (FE). These common behaviours illustrate the interaction patterns of learners in the virtual museum, and demonstrate the completeness of the system design, ensuring that both groups of students engaged with the same learning content. Specifically, Start Experience (SE) → Explore Object (EO) → Read Materials (RM) → Answer the question correctly (AC) → Explore Object (EO) highlights the process of exploring objects, acquiring detailed information, checking knowledge, and re-stimulating new exploration through successful feedback from the beginning of the experience. This process demonstrates the system's content design and checking mechanism, which effectively promotes the acquisition and consolidation of knowledge. Next Scene (NS) → Explore Object (EO) → Answer the question correctly (AC) indicates that learners could quickly engage in object exploration after switching to a new scene, and then promptly complete the knowledge check.

Besides, the red line in Figure 8 presents the main differences in the learning behaviours of the two groups. Students in the experimental group had a very high probability of repeatedly reading the materials after reading them once (Read Materials (RM) → Read Materials (RM)). This suggests that the SE-VML mode motivated learners to engage in repeated iterative message processing when comprehending the content, in order to fulfil the need for self-explanation of textual representations. Furthermore, the high probability that students in the experimental group moved to the next scene and continued to read the materials (Read Materials (RM) → Next Scene (NS) → Read Materials (RM)) suggests that the SE-VML model encouraged learners to proactively and purposefully move to a new scene after completing the self-explanation task in the current scene, and to prioritise searching for relevant content in preparation for the next explanatory task.

5.6 | Interview Results

Two researchers coded the interview data and Cohen's kappa was calculated at 0.74. A total of 214 codes were identified during the coding process, of which 152 were from the experimental group and 62 from the control group. In the end, the data from the experimental group were categorised into five themes comprising 16 subthemes, while the data from the control group were categorised into five themes comprising nine subthemes.

As can be seen in Figures 10 and 11, we distinguished the text data into application codes (green and red) and abstract codes (blue), and used the code-code relation of "is part of" to establish the links between them. In addition, the five abstract codes were connected to each other through the code-code relation of "is associated with," presenting the relatedness between the abstract codes. In response to research question 5, five themes were finalised.

1. Promoting Learning Effectiveness

Students mentioned several advantages that promote learning effectiveness. These included focusing on understanding the learning content, reviewing and reflecting on the learning content, and enhancing interdisciplinary knowledge. Besides, the experimental group specifically reported one

benefit: strengthening impressive understanding of knowledge (Table 8).

For example, E04 mentioned that by completing the self-explanation tasks guided by the structured questions, they were able to focus on understanding the learning content and grasp the learning focus. C02 stated that by exploring the knowledge background of the artefacts, they learned a great deal of interdisciplinary knowledge. In addition, E02 noted that the SE-VML approach helped them to review and reflect on the learning content repeatedly, which strengthened their understanding of the knowledge content and left a deep and impressive memory.

2. Fostering Critical Thinking

Students reported that the learning process fostered critical thinking. This included the ability to promote diverse thinking. Besides, the experimental group uniquely reported two abilities: evaluating the accuracy of viewpoints and developing independent thinking skills (Table 8). Most students in the experimental group stated that before participating in facilitating collaboration, they built their understanding and perspectives on the content through the process of self-explanation, which involved developing independent thinking skills. This process helped them more effectively evaluate the accuracy of viewpoints during group collaboration and provide constructive feedback.

TABLE 8 | Frequency of students' responses in the interviews.

Aspect	Item	Experimental group frequency	Control group frequency
Promoting learning effectiveness	Focus on understanding the learning content	10	6
	Review and reflect on the learning content	12	7
	Enhance interdisciplinary knowledge	11	9
	Strengthen impressive understanding of knowledge	10	0
Fostering critical thinking	Promote diverse thinking	9	5
	Evaluate the accuracy of viewpoints	9	0
	Develop independent thinking skills	8	0
Fostering collaboration	Communicate and share opinions	10	7
	Collaborate to complete tasks	8	6
	Achieve group consensus	8	0
Enhancing metacognition	Develop and refine conceptual knowledge	11	5
	Self-monitor comprehension	10	0
	Adapt learning strategies	9	0
Increasing learning motivation	Apply the approach to other courses	10	8
	Continue to use the approach	10	9
	Explore the learning content further	9	0

For example, E01 mentioned that after completing the self-explanation task, he was able to understand the content more clearly and could judge and distinguish the correctness of different opinions, providing relevant responses in communicating and sharing opinions during group discussions. C08 also noted that through this learning experience, he not only learned a great deal of interdisciplinary knowledge behind the artefacts, but also further broadened his thinking as it promoted diverse thinking.

3. Fostering Collaboration

In terms of facilitating collaboration, students mentioned several aspects. These included the ability to communicate and share opinions, as well as the ability to collaborate to complete tasks. Besides, the experimental group additionally reported an extra ability: achieving group consensus (Table 8). Most students in the experimental group believed that after completing the self-explanation task, they could express their viewpoints more clearly when entering group discussions. Through communicating and sharing opinions, they were able to compare and integrate different ideas, which helped them achieve group consensus. In contrast, students in the control group often mentioned situations where they either omitted certain content or struggled to explain their thoughts clearly.

For example, E09 mentioned that the atmosphere during group discussion was lively, with many members actively sharing their organised content and opinions. Through this process, the group was able to achieve group consensus and collaboratively complete the concept mapping task. C06 shared that during their group discussion, almost everyone forgot an important piece of background knowledge, but one classmate reminded the group, successfully avoiding the omission of critical information. C04 expressed gratitude to a group member who helped supplement and clarify their ideas when they struggled to explain their point of view. However, C04 also mentioned that some members mostly listened passively to others' opinions and only took action when asked to assist with minor tasks.

4. Enhancing Metacognition

In terms of enhancing metacognition, students mentioned the ability to develop and refine conceptual knowledge. Besides, the experimental group solely reported two abilities: self-monitoring comprehension and adapting learning strategies (Table 8). Most students in the experimental group stated that when answering the self-explanation task questions, they were better able to monitor their understanding of the learning content. They also became more aware of the information easily overlooked during the exploration of the virtual museum scenes, prompting them to adapt their learning strategies and focus more on the key information.

For example, E07 mentioned that by answering the structured guiding questions in the self-explanation task, they could clearly recognise their level of comprehension and tried to answer the questions in their own words, which helped build a stronger understanding of the knowledge. E05 reported that they initially thought they were already familiar with the learning content in the scene, but when completing the self-explanation task, they

realised there were still parts of the information they did not fully understand. Therefore, they went back to the scene to carefully review the explanations and then proceeded to the next scene to continue fulfilling the requirements of the self-explanation task. C03 shared that during the group discussion for creating the collaborative concept mapping, they realised they did not fully understand a specific piece of information. However, after listening to their group members' viewpoints, they were able to grasp the knowledge completely.

5. Increasing Learning Motivation

In terms of increasing learning motivation, students reported two common willingness indicators: their willingness to apply the approach to other courses and their willingness to continue to use the approach. Besides, the experimental group exclusively reported an additional aspect: the desire to explore the learning content further (Table 8). E04 stated that this learning approach sparked their interest in learning and made them curious about the interdisciplinary knowledge embedded in the artefacts. It also motivated them to proactively read the learning materials first whenever entering a new scene. C05 expressed that the learning method was very interesting, and they hoped to continue using it and apply it to different course content. Besides, E08 mentioned that the museum collections contain many artefacts, but this learning activity only focused on a few items. After completing the collaborative concept mapping with the group, E08 expressed the hope to have more opportunities to explore additional artefacts in the future.

6 | Discussion and Conclusions

6.1 | Discussion

The results showed that, compared with the C-VML group, students in the SE-VML group obtained significantly higher post-test scores in learning achievement, critical thinking tendency, collaboration tendency, and metacognition. In addition, behavioural log analysis and interview data revealed clear differences in the learning processes and perceived experiences between the two conditions.

The findings from the statistical analyses, behavioural sequence analysis, and interview data collectively provide convergent and triangulated evidence supporting the effectiveness of the SE-VML model. Specifically, the superior performance of the experimental group is substantiated by behavioural evidence indicating more frequent rereading and goal-oriented exploration, which reflect deeper cognitive engagement and more purposeful information processing. Furthermore, the interview findings offer complementary explanatory insights, showing that learners perceived self-explanation as beneficial for organising knowledge, clarifying understanding, and preparing for group discussion. Taken together, these results suggest that self-explanation enhances individual cognitive processing and facilitates the transition from individual knowledge construction to effective collaborative interaction.

From the perspective of the constructivist 4C framework, these findings can be interpreted through the cognitive mechanism

underlying the self-explanation process. By requiring learners to externalise and organise their understanding prior to collaboration, the SE-VML model strengthens the construction stage and enhances learners' cognitive readiness for subsequent conversation and collaboration stages (Feyzi Behnagh and Yasrebi 2020; Lawson and Mayer 2021; Murphy 1997). Consequently, learners enter social interaction with more coherent ideas and greater confidence, enabling more effective knowledge sharing and co-construction (Hsu et al. 2016; Petersen et al. 2023; Schnaubert and Bodemer 2022).

1. Learning Achievement

Students in the experimental group performed significantly better than those in the control group. This result is consistent with previous studies demonstrating that the SE-VML model, which requires learners to engage in self-explanation, can enhance learning achievement, as reported in prior studies (Bisra et al. 2018; Fiorella et al. 2020; Wang et al. 2023). As self-explanation facilitates the operation and encoding of working memory, learners are able to more effectively process learning materials in depth. The ability to maintain and encode working memory affects long-term memory and influences the effectiveness of learning (Lachner et al. 2021; Pi et al. 2021). In addition, the process of self-explanation enhances internal attention and filters out irrelevant information through sensory inhibition, allowing learners to focus on the content (Pi et al. 2021). When learners find that they cannot clearly explain a concept, they may review the content and reorganise the information. Such a process can lead to a generation effect, which helps learners build a more solid knowledge structure and is a key factor in improving learning achievement (Endres et al. 2024; Fukaya 2013).

Within the 4C framework, self-explanation supported the construction stage by externalising personal understandings into written representations (Feyzi Behnagh and Yasrebi 2020; Lawson and Mayer 2021; Murphy 1997). Chi and Wylie (2014) showed that learning achievement increases with the increase of cognitive engagement. Especially through written generation and externalisation, learners trigger deeper cognitive processing. Compared with remaining internal, written self-explanation turns vague, fragmented ideas into concrete, logically structured prose. Parong (2022) showed that, in VR, generative learning strategies such as written self-explanations improve learning outcomes. Parong and Mayer (2018) also found that written self-explanations in VR enhance learning and deepen understanding of complex knowledge. A written self-explanation creates space to revisit, revise, and refine arguments. Thus, written self-explanation functions not only as a record of thought but also as a structured tool for cognitive processing.

The written outputs of self-explanation supported the conversation and collaboration stages by providing shareable materials for social interaction and knowledge co-construction (Hsu et al. 2016; Petersen et al. 2023; Schnaubert and Bodemer 2022). Students in the experimental group were involved in multiple levels of cognitive engagement during the learning process, including the active mode (deepening understanding through repeated retrieval of virtual museum information), the constructive mode (generating explanations of knowledge concepts with words and organising learning content with group

members using concept maps), and the interactive mode (exchanging ideas and collaborating on concept mapping with members). In particular, students in the experimental group had more cognitive involvement in the constructive mode through self-explanation than students in the control group (Lawson and Mayer 2021). In the interview, E02 reported that completing the self-explanation task and then completing the concept map with their group members could repeatedly deepen the impression of the learning content.

2. Critical Thinking Tendency

Students in the experimental group performed significantly better than those in the control group. This result suggests that through the SE-VML model, students were able to actively construct knowledge and deepen their understanding through logical reasoning, conceptual integration, and argumentation, rather than passively accepting information. When students represent learning content in their own way, attempt to connect new information with prior knowledge, and make inferences or bridging speculations, they engage in meaningful cognitive processing that lays the foundation for analytical and evaluative thinking (Fiorella and Mayer 2016; Hand et al. 2021).

Although the self-explanation questions primarily focused on conceptual understanding and knowledge connection, this process functioned as a cognitive preparation phase that enabled learners to organise ideas, recognise inconsistencies, and develop coherent mental models (Lawson and Mayer 2021; Wen et al. 2024). This finding shows that self-explanation supported the construction stage by transforming raw information into personally meaningful knowledge representations (Fiorella 2023; Wen et al. 2024). Such cognitive activities are essential prerequisites to critical thinking, as they help learners clarify what they know before engaging in collaborative reasoning (Lachner et al. 2021). As several scholars have indicated, self-explanation can encourage learners to monitor comprehension, construct arguments, and reflect on their reasoning processes (Fajrin and Pusparini 2019; Wen et al. 2024), which could support higher-level thinking during subsequent discussions.

Critical thinking is an active and organised cognitive process of becoming aware of and reflecting on one's own thought processes in order to enhance one's understanding and analyse what is going on in the surrounding environment. It also involves evaluating one's own and others' perspectives to make reasoned judgements (Danczak et al. 2020; Tuaputty et al. 2021). While the SE-VML model did not directly induce critical thinking through self-explanation alone, it helped guide students to engage more effectively in the subsequent collaborative phase. This supported the conversation and collaboration stages of the 4C framework by providing shareable cognitive materials for group discussion and knowledge co-construction (Hsu et al. 2016; Petersen et al. 2023; Schnaubert and Bodemer 2022). Having learners externalise their understanding before discussion enabled them to participate in group dialogue with clearer personal perspectives (Hu and Hwang 2024; Ting et al. 2018). When students are cognitively prepared, they can more effectively compare, question, and negotiate different viewpoints, thereby exercising evaluative and analytical thinking (Danczak et al. 2020; Tuaputty et al. 2021).

Students in the SE-VML group began the learning process by organising individuals' knowledge through self-explanation to ensure that each learner would develop a personal view of what they had learned before entering the group discussions, thus reducing reliance on other members' ideas (Ting et al. 2018). This mode of learning strengthened individual cognitive preparation and allowed learners to more actively participate, question, and integrate different viewpoints through critical reasoning during group collaboration (Danczak et al. 2020; Tuaputty et al. 2021; Wen et al. 2024). Interview data support this interpretation. For example, participant E01 mentioned that after completing the self-explanation task, they became more aware of what they had learned and were better able to evaluate the accuracy of others' ideas during group discussions. This finding illustrates how self-explanation, while not directly eliciting critical thinking, sets the cognitive stage for it to occur during collaborative knowledge construction.

3. Collaborative Tendencies

Students in the experimental group performed significantly better than those in the control group, a result that underscores the critical role of self-explanation in collaborative tendencies. Ting et al. (2018) used self-explanation-based guiding questions to help students externalise their internal thinking, which in turn facilitated cognitive conflict and conceptual adjustment during group collaboration, and the students gradually developed a common understanding to achieve knowledge sharing. This study is consistent with previous research showing that generating personalised understanding through self-explanation enables students to participate more effectively in discussions during collaboration and to compare and contrast reasoning when communicating with partners (Hsu et al. 2016). Students were able to organise their thoughts more clearly when they first engaged in self-explanation for a more structured presentation (Pi et al. 2022). Interview results also showed that students in the experimental group felt that they were able to express their views more clearly during group discussions, whereas students in the control group mentioned that they had vaguer memory of artefact details, and were unable to organise and explain their ideas in a structured way. From a cognitive view, self-explanation builds cognitive readiness (Fiorella 2023). Hand et al. (2021) showed that when learners turn scattered information into structured personal knowledge, they enter collaboration with usable cognitive capital. They are no longer passive recipients. They contribute ideas, ask questions, and engage in meaning negotiation. From a motivational viewpoint, being prepared strengthens perceived competence. This feeling drives a collaborative tendency (Schnaubert and Bodemer 2022).

Within the 4C framework, self-explanation first supported the construction stage by helping learners turn personal understandings into written representations. These written outputs then fed into the conversation and collaboration stages (Lachner et al. 2021; Lawson and Mayer 2021). Liang et al. (2021) demonstrated that the collaborative process, whereby each student was actively involved in the learning process by expressing their personal views, and was integrated into the team discussion and decision-making, helped with knowledge sharing and collaborative tendency. In addition, Chi and Wylie (2014) mentioned

that the key to the interactive model is collaboration to create new knowledge rather than unidirectional transmission of information. In the interview results, students in the experimental group mentioned that they felt that they and their group members were able to participate more actively in the discussion, respond positively to their peers' comments, and work together to complete the knowledge concept map. In contrast, students in the control group mentioned that some of the students in the group only passively listened to their peers and waited for tasks to be assigned, or did not even participate actively. In this case, the control group exhibited the passive mode of learning mentioned by Chi and Wylie (2014), as they only received information without active participation. Therefore, when students organise their knowledge at the individual level through self-explanation before collaborative learning, it can make them more engaged in the collaborative process and enhance their collaborative tendency.

4. Metacognition

Students in the experimental group performed significantly better than those in the control group in terms of metacognition. This result echoes the SE-VML model, in which self-explanation helps learners to monitor their own understanding of learning, and effectively enhances their metacognition (Chen et al. 2025). In the process of self-explanation, learners need to link new information to existing knowledge (i.e., encoding) and express their understanding through text, language, or other means (i.e., retrieval) in order to review and strengthen the knowledge structure (Endres et al. 2024; Fukaya 2013; Lachner et al. 2021). This process not only enhances learners' awareness of their own state of understanding, but also prompts them to proactively correct possible cognitive errors or conceptual omissions. Similarly, in concept mapping, learners need to analyse and organise the conceptual associations (coding encoding) among the learning materials, and then visualise the structured knowledge (extraction) to facilitate the integration and application of knowledge (Endres et al. 2024; Fukaya 2013). In addition, the combination of self-explanation and collaborative concept mapping strategies not only helps learners to systematically organise information, but also enhances their metacognition through the interactive mechanism of encoding and extraction. This is because learners need to constantly retrieve and construct situational models to make it easier for them to monitor the state of their understanding (Endres et al. 2024; Fukaya 2013). When learners keep everything in their heads, they often overestimate their understanding. Only when they are asked to write it out do gaps and incoherence surface (Hand et al. 2021). Fukaya (2013) found that a written-explanation task compels learners to think about their thinking—the essence of metacognitive monitoring.

Within the 4C framework, self-explanation supported the construction stage by externalising thinking into written form. This served as a metacognitive checkpoint that helped learners detect gaps in their understanding (Endres et al. 2024; Lawson and Mayer 2021). Therefore, self-explanation is indeed effective in terms of triggering metacognitive processes, prompting learners to develop more effective cognitive strategies, externalise their knowledge, and continually monitor and adjust their level of understanding (Fiorella and Mayer 2016; Roscoe and Chi 2008). The interview results also showed that most of the students in

the experimental group mentioned that they were able to recognise their level of understanding during the activities and adjusted their learning strategies according to their deficiencies in order to strengthen their understanding of the knowledge concepts. Therefore, the SE-VML model can indeed enhance learners' metacognition.

5. Learning Behaviour and Experience

In terms of learning behaviour analysis and perceptual experience, students in the experimental group significantly repeated reading the materials after reading them once (Read Materials (RM) → Read Materials (RM)). This may be due to the task of self-explanation, which prompted them to absorb information repeatedly and carefully. Elme et al. (2022) mentioned that learners need to comprehend the content accurately in order to complete the self-explanation of the textual representations, and Buchner (2021) pointed out that the initial comprehension of learners is often not sufficient for clear and complete representation, which may be due to vague memories or the need for precise vocabulary to support the information. In the interview, E02 also pointed out that this mode of learning facilitates repeated review and reflection. In contrast, the control students were not tasked with self-explanation, and so were less motivated to repeat reading as they required less text organisation and comprehension in the learning process. Therefore, the control group students were less likely to reread the materials.

In addition, after reading the materials, students in the experimental group moved to the next scene and continued to read the materials (Read Materials (RM) → Next Scene (NS) → Read Materials (RM)). This may be because after reading the materials and completing the self-explanation task for the corresponding scene, they were motivated by the next self-explanation task to move to the new scene and prioritise the reading of the learning resources in order to collect the required content knowledge. In the interviews, E05 indicated that when attempting to complete the self-explanation task, students read the materials again when they found that they did not have sufficient understanding of the content, and immediately moved on to the next scene to continue to read the content carefully. E04 reported that this mode of learning stimulated their interest in learning and prompted them to actively read the learning resources after entering a new scenario. In contrast, although students in the control group also changed scenes, they were motivated by the completion of the exploration in the scene or their personal interests, and lacked the clear milestones and reading motivation brought by the self-explanation tasks. Therefore, the Read Materials (RM) → Next Scene (NS) → Read Materials (RM) behaviour of the experimental group helps to present the structured cognitive processing history of the SE-VML model (Wang et al. 2023; Wen et al. 2024). Within the 4C framework, self-explanation drove repeated reading and organised information processing. The shift from reading to scene transition marks the cognitive shift from exploring the context to constructing personal knowledge.

On the other hand, the experimental group shared the same salient behaviours with the control group, including the Start Experience (SE) → Explore Object (EO) → Read Materials (RM) → Answer the question correctly (AC) → Explore Object

(EO), Next Scene (NS) → Explore Object (EO) → Answer the question correctly (AC), and Explore Object (EO) → Finish Experience (FE) behavioural paths. These common behavioural paths emerged because the learning design was grounded in the 4C framework. These paths show that the learners had the same basic interaction patterns when exploring information and constructing knowledge in the virtual museum. They also revealed the structural consistency of the design of the virtual museum system, which can guide two groups of students with different learning methods to invest the same resources in their learning. Specifically, the cycle of exploration and knowledge construction is clearly visible: Start Experience (SE) → Explore Object (EO) → Read Materials (RM) → Answer the question correctly (AC) → Explore Object (EO). This path describes the learner's experience from the beginning, exploring the objects, then reading the materials for information and answering the questions correctly. As mentioned by Hwang et al. (2023), the positive feedback of successful answers motivates learners to continue exploring, creating a virtuous learning cycle of exploration and comprehension. In the interview, E08 indicated that he had just entered the virtual museum and found the related educational materials by exploring the objects. After answering the questions correctly, he was more interested in exploring other objects.

In addition, the rapid path of knowledge checking after the scene change was also evident: Next Scene (NS) → Explore Object (EO) → Answer the question correctly (AC), which indicated that learners explored the objects quickly and then answered the related questions correctly after entering the new scene, showing the close connection between exploration and knowledge checking. C02 indicated that after entering a new scene, they were able to gather information and answer the questions correctly by exploring the objects. Meanwhile, Explore Object (EO) → Finish Experience (FE) indicates that learners could end the experience in an organised manner after exploring the objects, reflecting the completion of the learning process in stages.

These common behavioural patterns and the learners' perceptual experiences not only indicate that the virtual museum has a good design for learning guidance and interaction, but also show that both groups of learners could experience the same learning process using different learning strategies. In addition, these same paths constitute the baseline of learner behaviours generated in the virtual museum, which makes the additional learning behaviours induced by the self-explanation task in the experimental group more recognisable and helps to present the learning behaviours generated by the SE-VML model.

6.2 | Limitations and Future Research

Several limitations of this study should be acknowledged. First, this study adopted a post-test-only design with a small sample size. While random assignment helps establish baseline comparability, it does not guarantee perfect equivalence in small samples. Future studies are encouraged to include larger samples and to conduct pre-tests. Second, learners did not receive immediate feedback during the written self-explanation task, unlike the real-time feedback provided for structured guiding questions. While this design was intended to promote individual

generative processing, the absence of immediate feedback may have delayed the correction of misconceptions. Future studies could integrate AI-based or adaptive feedback mechanisms to examine how different feedback timings influence learning processes and outcomes.

Third, the study focused on virtual museum learning by elementary school students, which may limit the generalisability of the findings. Although this context was theoretically and practically relevant, future studies should examine the applicability of the SE-VML model across different learning contexts (e.g., flipped or blended learning), technological environments (e.g., augmented reality, generative artificial intelligence, or metaverse platforms), disciplines, and educational levels. In addition, detailed SES data were not available in this school-based context; therefore, potential SES-related influences could not be directly examined.

Fourth, while this study examined learning achievement, critical thinking tendency, collaboration tendency, and metacognition, other important variables, such as cognitive load and domain-specific higher-order thinking skills (e.g., problem solving), were not directly measured. Future research should incorporate these variables to further validate and refine the instructional design.

Finally, although the findings suggest differences in collaborative tendencies and learning processes between conditions, the present analyses did not capture fine-grained interaction dynamics within group collaboration. Future studies could employ epistemic network analysis to examine interaction patterns and cognitive connections during collaborative knowledge construction, thereby providing a more comprehensive understanding of how the SE-VML model shapes learning processes.

6.3 | Conclusions

Overall, this study demonstrated the feasibility of integrating self-explanation into virtual reality. Using a randomised controlled design, we systematically compared a self-explanation-guided virtual museum learning (SE-VML) model with a conventional virtual museum learning (C-VML) model. A triangulated analysis, combining questionnaires, backend behavioural logs, and qualitative interviews, converged on a clear conclusion: in virtual museum learning, adding a structured, explicit self-explanation task is an effective instructional strategy that significantly strengthens learners' multidimensional cognitive abilities.

In concrete terms, the SE-VML model not only improved learning achievement but also outperformed the C-VML model in critical thinking tendency, collaborative tendency, and metacognition. In the 4C framework, self-explanation supported construction through written externalisation, and these written outputs fed into the conversation and collaboration stages. Behavioural analyses and interview data clarified the mechanism: the self-explanation task prompted cognitive processing, for example rereading and written organisation, which supported personal knowledge construction and reflective critique, and provided the cognitive readiness needed for subsequent collaborative discussion.

From a theoretical standpoint, these findings offer empirical support for applying self-explanation in virtual environments. Written externalisation functioned as a cognitive tool that activated metacognitive monitoring, raised achievement, and, in later collaboration, promoted critical thinking and collaborative tendency. The SE-VML sequence, individual first and collaboration next, validated an effective way to integrate individual cognitive deepening with the social co-construction of knowledge.

From a practical standpoint, the results provide a concrete and effective design model for virtual museums and other digital learning contexts. To foster deep learning, instructors should embed guided scaffolds that prompt individual knowledge construction and reflection. Such scaffolds would not only enhance individual outcomes but also lay a foundation for productive teamwork, mitigating common issues such as shallow discussion and social loafing. In short, this study verifies an innovative learning model and offers guidance for designing activities that support both individual depth and team efficiency in increasingly prevalent virtual settings.

Author Contributions

Gwo-Jen Hwang: conceptualization, supervision, validation, investigation, methodology, writing – review and editing. **Yi-Chun Lu:** conceptualization, methodology, data curation, investigation, writing – original draft, formal analysis. **Yun-Fang Tu:** writing – review and editing, validation, visualization, supervision, conceptualization, methodology, investigation.

Funding

This work was supported in part by the National Science and Technology Council of Taiwan under contract numbers NSTC 112-2410-H-011-012-MY3 and NSTC 113-2410-H-142-004-MY3. The study is also supported by the “Empower Vocational Education Research Center” of National Taiwan University of Science and Technology (NTUST) from the Featured Areas Research Center Programme within the framework of the Higher Education Sprout Project by the Ministry of Education (MOE) in Taiwan.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Peer Review

For transparency, the peer review documents associated with this article are available at <https://doi.org/10.1002/jcal.70296>.

References

- Baloche, L., and C. M. Brody. 2017. “Cooperative Learning: Exploring Challenges, Crafting Innovations.” *Journal of Education for Teaching* 43, no. 3: 274–283. <https://doi.org/10.1080/02607476.2017.1319513>.
- Bisra, K., Q. Liu, J. C. Nesbit, F. Salimi, and P. H. Winne. 2018. “Inducing Self-Explanation: A Meta-Analysis.” *Educational Psychology Review* 30, no. 3: 703–725. <https://doi.org/10.1007/s10648-018-9434-x>.

- Buchner, J. 2021. "Generative Learning Strategies Do Not Diminish Primary Students' Attitudes Towards Augmented Reality." *Education and Information Technologies* 27, no. 1: 701–717. <https://doi.org/10.1007/s10639-021-10445-y>.
- Chai, C. S., F. Deng, P. S. Tsai, J. H. L. Koh, and C. C. Tsai. 2015. "Assessing Multidimensional Students' Perceptions of Twenty-First-Century Learning Practices." *Asia Pacific Education Review* 16, no. 3: 389–398. <https://doi.org/10.1007/s12564-015-9379-4>.
- Chen, W., L. Lin, and W. Pang. 2025. "Learning via Video: The Effects of Summarizing Strategies and Self-Explanation Prompts." *Education and Information Technologies* 30, no. 3: 3541–3572. <https://doi.org/10.1007/s10639-024-12915-5>.
- Chi, M. T. H., M. Bassok, M. W. Lewis, P. Reimann, and R. Glaser. 1989. "Self-Explanations: How Students Study and Use Examples in Learning to Solve Problems." *Cognitive Science* 13, no. 2: 145–182. https://doi.org/10.1207/s15516709cog1302_1.
- Chi, M. T. H., and R. Wylie. 2014. "The ICAP Framework: Linking Cognitive Engagement to Active Learning Outcomes." *Educational Psychologist* 49, no. 4: 219–243. <https://doi.org/10.1080/00461520.2014.965823>.
- Chiu, M. C., G. J. Hwang, and L. H. Hsia. 2023. "Promoting Students' Artwork Appreciation: An Experiential Learning-Based Virtual Reality Approach." *British Journal of Educational Technology* 54, no. 2: 603–621. <https://doi.org/10.1111/bjet.13265>.
- Cohen, J. 1988. *Statistical Power Analysis for the Behavioral Sciences*. 2nd ed. Erlbaum.
- Creswell, J. W., and C. N. Poth. 2016. *Qualitative Inquiry and Research Design: Choosing Among Five Approaches*. Sage Publications. <https://doi.org/10.1177/1524839915580941>.
- Danczak, S. M., C. D. Thompson, and T. L. Overton. 2020. "Development and Validation of an Instrument to Measure Undergraduate Chemistry Students' Critical Thinking Skills." *Chemistry Education Research and Practice* 21, no. 1: 62–78. <https://doi.org/10.1039/c8rp00130h>.
- De Back, T. T., A. M. Tinga, and M. M. Louwerse. 2023. "Learning in Immersed Collaborative Virtual Environments: Design and Implementation." *Interactive Learning Environments* 31, no. 8: 5364–5382. <https://doi.org/10.1080/10494820.2021.2006238>.
- Elme, L., M. L. Jørgensen, G. Dandanell, A. Mottelson, and G. Makransky. 2022. "Immersive Virtual Reality in STEM: Is IVR an Effective Learning Medium and Does Adding Self-Explanation After a Lesson Improve Learning Outcomes?" *Educational Technology Research and Development: ETR & D* 70, no. 5: 1601–1626.
- Endres, T., S. Carpenter, and A. Renkl. 2024. "Constructive Retrieval: Benefits for Learning, Motivation, and Metacognitive Monitoring." *Learning and Instruction* 94: 101974. <https://doi.org/10.1016/j.learninstruc.2024.101974>.
- Fajrin, R. N., and R. Pusparini. 2019. "Self-Explanation Strategy: Supporting Students Critical Thinking in Reading Comprehension for Science Class in Senior High School." *RETAIN: Journal of Research in English Language Teaching* 7, no. 1: 98–106.
- Feyzi Behnagh, R., and S. Yasrebi. 2020. "An Examination of Constructivist Educational Technologies: Key Affordances and Conditions." *British Journal of Educational Technology* 51, no. 6: 1907–1919. <https://doi.org/10.1111/bjet.13036>.
- Fiorella, L. 2023. "Making Sense of Generative Learning." *Educational Psychology Review* 35, no. 2: 50. <https://doi.org/10.1007/s10648-023-09769-7>.
- Fiorella, L., and R. E. Mayer. 2016. "Eight Ways to Promote Generative Learning." *Educational Psychology Review* 28, no. 4: 717–741. <https://doi.org/10.1007/s10648-015-9348-9>.
- Fiorella, L., A. T. Stull, S. Kuhlmann, and R. E. Mayer. 2020. "Fostering Generative Learning From Video Lessons: Benefits of Instructor-Generated Drawings and Learner-Generated Explanations." *Journal of Educational Psychology* 112, no. 5: 895–906. <https://doi.org/10.1037/edu0000408>.
- Fukaya, T. 2013. "Explanation Generation, Not Explanation Expectancy, Improves Metacomprehension Accuracy." *Metacognition and Learning* 8, no. 1: 1–18. <https://doi.org/10.1007/s11409-012-9093-0>.
- Hand, B., Y. C. Chen, and J. K. Suh. 2021. "Does a Knowledge Generation Approach to Learning Benefit Students? A Systematic Review of Research on the Science Writing Heuristic Approach." *Educational Psychology Review* 33, no. 2: 535–577. <https://doi.org/10.1007/s10648-020-09550-0>.
- Ho, C. M. L., M. E. Nelson, and W. Müller-Wittig. 2011. "Design and Implementation of a Student-Generated Virtual Museum in a Language Curriculum to Enhance Collaborative Multimodal Meaning-Making." *Computers & Education* 57, no. 1: 1083–1097. <https://doi.org/10.1016/j.compedu.2010.12.003>.
- Hsu, C. Y., C. C. Tsai, and H. Y. Wang. 2016. "Exploring the Effects of Integrating Self-Explanation Into a Multi-User Game on the Acquisition of Scientific Concepts." *Interactive Learning Environments* 24, no. 4: 844–858. <https://doi.org/10.1080/10494820.2014.926276>.
- Hu, Y., and G. J. Hwang. 2024. "Promoting Students' Higher Order Thinking in Virtual Museum Contexts: A Self-Adapted Mobile Concept Mapping-Based Problem Posing Approach." *Education and Information Technologies* 29, no. 3: 2741–2765. <https://doi.org/10.1007/s10639-023-11930-2>.
- Hwang, G. J., H. C. Chen, C. Y. Hsu, and G. H. Hwang. 2023. "Effects of a Graphic Organizer-Based Two-Tier Test Approach on Students' Learning Achievement and Behaviors in Spherical Video-Based Virtual Learning Contexts." *Computers & Education* 198: 104757. <https://doi.org/10.1016/j.compedu.2023.104757>.
- Hwang, G. J., T. C. Yang, C. C. Tsai, and S. J. Yang. 2009. "A Context-Aware Ubiquitous Learning Environment for Conducting Complex Science Experiments." *Computers & Education* 53, no. 2: 402–413. <https://doi.org/10.1016/j.compedu.2009.02.016>.
- Islek, D., and I. Danju. 2019. "The Effect of Museum Education Practices Carried Out on Virtual Teaching Environments on Prospective Teachers' Views." *Revista de Cercetare Si Interventie Sociala* 67: 114–135. <https://doi.org/10.33788/rcis.67.8>.
- Jacob, L., A. Lachner, and K. Scheiter. 2020. "Learning by Explaining Orally or in Written Form? Text Complexity Matters." *Learning and Instruction* 68: 101344. <https://doi.org/10.1016/j.learninstruc.2020.101344>.
- Jia, Y., X. E. Wang, Z. P. T. Sin, et al. 2024. "Knowledge-Graph-Driven Mind Mapping for Immersive Collaborative Learning: A Pilot Study in Edu-Metaverse." *IEEE Transactions on Learning Technologies* 17: 1794–1808. <https://doi.org/10.1109/tlt.2024.3406638>.
- Karakuş Yılmaz, T., Z. Yılmaz, and Ö. Arpacık. 2025. "Gamified Mobile Guidance for Museum Virtual Tours: How Task Design Shapes Learning Experience and Cognitive Load." *Journal of Educational Computing Research* 63, no. 6: 1430–1459. <https://doi.org/10.1177/07356331251347008>.
- Kayaalp, F., Z. B. Namlı, and E. Meral. 2024. "My Museum: A Study of Pre-Service Social Studies Teachers' Experience in Designing Virtual Museums." *Education and Information Technologies* 29, no. 18: 24047–24085. <https://doi.org/10.1007/s10639-024-12742-8>.
- Keppel, G., and T. D. Wickens. 2004. *Design and Analysis: A Researcher's Handbook*. Prentice Hall.
- Kim, S. J., and Y. M. Ju. 2023. "A Study on the Design of Smart Tourism Concept Map Based on the Model of Advance Organizer That Attracts Interest for Space Telling in Metaverse." *Korean Institute of Smart Media* 12, no. 8: 45–59. <https://doi.org/10.30693/smj.2023.12.8.45>.
- Lachner, A., L. Jacob, and V. Hoogerheide. 2021. "Learning by Writing Explanations: Is Explaining to a Fictitious Student More Effective Than

- Self-Explaining?" *Learning and Instruction* 74: 101438. <https://doi.org/10.1016/j.learninstruc.2020.101438>.
- Lai, C. L., and G. J. Hwang. 2014. "Effects of Mobile Learning Time on Students' Conception of Collaboration, Communication, Complex Problem-Solving, Meta-Cognitive Awareness, and Creativity." *International Journal of Mobile Learning and Organisation* 8, no. 3: 276–291. <https://doi.org/10.1504/IJMLO.2014.067029>.
- Lawson, A. P., and R. E. Mayer. 2021. "Benefits of Writing an Explanation During Pauses in Multimedia Lessons." *Educational Psychology Review* 33, no. 4: 1859–1885. <https://doi.org/10.1007/s10648-021-09594-w>.
- Lee, H., T. H. Jung, M. C. tom Dieck, and N. Chung. 2020. "Experiencing Immersive Virtual Reality in Museums." *Information & Management* 57, no. 5: 103229. <https://doi.org/10.1016/j.im.2019.103229>.
- Liang, H. Y., T. Y. Hsu, G. J. Hwang, S. C. Chang, and H. C. Chu. 2021. "A Mandatory Contribution-Based Collaborative Gaming Approach to Enhancing Students' Collaborative Learning Outcomes in Science Museums." *Interactive Learning Environments* 31, no. 5: 2692–2706. <https://doi.org/10.1080/10494820.2021.1897845>.
- Lin, C. J., K. Y. Tang, and Y. F. Tu. 2023. "Advancements and Research Topics of Museum-Based Mobile Learning: A Review of Journal Publications From 2008 to 2019." *Interactive Learning Environments* 31, no. 8: 4729–4752. <https://doi.org/10.1080/10494820.2021.1981394>.
- Lin, H. C., G. J. Hwang, and Y. D. Hsu. 2019. "Effects of ASQ-Based Flipped Learning on Nurse Practitioner Learners' Nursing Skills, Learning Achievement and Learning Perceptions." *Computers & Education* 139: 207–221. <https://doi.org/10.1016/j.compedu.2019.05.014>.
- Makransky, G. 2022. "The Immersion Principle in Multimedia Learning." In *The Cambridge Handbook of Multimedia Learning*, edited by R. E. Mayer and L. Fiorella, 3rd ed., 296–303. Cambridge University Press.
- Mayer, R. E., G. Makransky, and J. Parong. 2022. "The Promise and Pitfalls of Learning in Immersive Virtual Reality." *International Journal of Human-Computer Interaction* 39, no. 11: 2229–2238. <https://doi.org/10.1080/10447318.2022.2108563>.
- Murphy, E. 1997. Constructivism: From Philosophy to Practice. <https://eric.ed.gov/?id=ED444966>.
- Parong, J. 2022. "Multimedia Learning in Virtual and Mixed Reality." In *The Cambridge Handbook of Multimedia Learning*, edited by R. E. Mayer and L. Fiorella, 3rd ed., 498–509. Cambridge University Press. <https://doi.org/10.1017/9781108894333.051>.
- Parong, J., and R. E. Mayer. 2018. "Learning Science in Immersive Virtual Reality." *Journal of Educational Psychology* 110, no. 6: 785–797. <https://doi.org/10.1037/edu0000241>.
- Petersen, G. B., V. Stenberdt, R. E. Mayer, and G. Makransky. 2023. "Collaborative Generative Learning Activities in Immersive Virtual Reality Increase Learning." *Computers & Education* 207: 104931. <https://doi.org/10.1016/j.compedu.2023.104931>.
- Pi, Z., Y. Zhang, D. Shi, X. Guo, and J. Yang. 2022. "Is Self-Explanation Better Than Explaining to a Fictitious Student When Learning From Video Lectures?" *British Journal of Educational Technology* 53, no. 6: 2012–2028. <https://doi.org/10.1111/bjet.13230>.
- Pi, Z., Y. Zhang, W. Zhou, et al. 2021. "Learning by Explaining to Oneself and a Peer Enhances Learners' Theta and Alpha Oscillations While Watching Video Lectures." *British Journal of Educational Technology* 52, no. 2: 659–679. <https://doi.org/10.1111/bjet.13048>.
- Puig, A., I. Rodríguez, J. L. Arcos, et al. 2020. "Lessons Learned From Supplementing Archaeological Museum Exhibitions With Virtual Reality." *Virtual Reality* 24, no. 2: 343–358. <https://doi.org/10.1007/s10055-019-00391-z>.
- Puntambekar, S. 2006. "Analyzing Collaborative Interactions: Divergence, Shared Understanding and Construction of Knowledge." *Computers & Education* 47, no. 3: 332–351. <https://doi.org/10.1016/j.compedu.2004.10.012>.
- Rajanaarivo, L., E. Maisel, and P. De Loor. 2019. "An Evolving Museum Metaphor Applied to Cultural Heritage for Personalized Content Delivery." *User Modeling and User-Adapted Interaction* 29, no. 1: 161–200. <https://doi.org/10.1007/s11257-019-09222-x>.
- Ratumbuisang, K. F., U. Hariyanti, G. E. Saputro, and Y. A. Abdilah. 2024. "Application of Collaborative Storytelling With Virtual Museum: Modeling the Relationship Among Pre-Service Teachers' Collaboration and Technology Acceptance." *Education and Information Technologies* 29, no. 16: 21943–21963. <https://doi.org/10.1007/s10639-024-12729-5>.
- Roscoe, R. D., and M. T. H. Chi. 2008. "Tutor Learning: The Role of Explaining and Responding to Questions." *Instructional Science* 36, no. 4: 321–350. <https://doi.org/10.1007/s11251-007-9034-5>.
- Roy, M., and M. T. H. Chi. 2005. "The Self-Explanation Principle in Multimedia Learning." In *The Cambridge Handbook of Multimedia Learning*, 271–286. Cambridge University Press. <https://doi.org/10.1017/cbo9780511816819.018>.
- Schnaubert, L., and D. Bodemer. 2022. "Group Awareness and Regulation in Computer-Supported Collaborative Learning." *International Journal of Computer-Supported Collaborative Learning* 17, no. 1: 11–38. <https://doi.org/10.1007/s11412-022-09361-1>.
- Schwartz, D. L., J. M. Tsang, and K. P. Blair. 2016. *The ABCs of How we Learn: 26 Scientifically Proven Approaches, How They Work, and When to Use Them*. W. W. Norton & Company.
- Seprum, P., and C. Wongwatkit. 2025. "Effects of an Immersive Virtual Environment-Based Gaming Approach on Students' Occupational Health Performance, Immersion and Motivation." *International Journal of Mobile Learning and Organisation* 19, no. 1: 33–58. <https://doi.org/10.1504/IJMLO.2025.142902>.
- Shehade, M., and T. Stylianou-Lambert. 2020. "Virtual Reality in Museums: Exploring the Experiences of Museum Professionals." *Applied Sciences* 10, no. 11: 4031. <https://doi.org/10.3390/app10114031>.
- Sweller, J. 2011. "Cognitive Load Theory." In *The Psychology of Learning and Motivation*, edited by J. P. Mestre and B. H. Ross, 37–76. Elsevier Academic Press.
- Tatlı, Z., G. Çelenk, and D. Altınışık. 2023. "Analysis of Virtual Museums in Terms of Design and Perception of Presence." *Education and Information Technologies* 28, no. 7: 8945–8973. <https://doi.org/10.1007/s10639-022-11561-z>.
- Ting, Y. L., Y. Tai, T. H. Tseng, and S. P. Tsai. 2018. "Innovative Use of Mobile Video Conferencing in Face-To-Face Collaborative Science Learning: The Case of Reflection in Optics." *Journal of Educational Technology & Society* 21, no. 3: 74–85. <https://www.jstor.org/stable/26458508>.
- Tu, Y. F. 2025. "Advancements and Paradigm Shifts in Mobile and Ubiquitous Learning: A Bibliometric Analysis of the 16 Years of the International Journal of Mobile Learning and Organisation." *International Journal of Mobile Learning and Organisation* 19, no. 1: 113–134. <https://doi.org/10.1504/IJMLO.2025.142915>.
- Tuaputty, H., M. Leasa, A. D. Corebima, and J. R. Batlolona. 2021. "The Correlation Between Critical Thinking Skills and Cognitive Learning Outcomes." *İlköğretim Online* 20, no. 1: 302–317. <https://doi.org/10.17051/ilkonline.2021.01.029>.
- van Peppen, L. M., P. P. J. L. Verhoeven, A. E. G. Heijltjes, E. M. Janssen, D. Koopmans, and T. van Gog. 2018. "Effects of Self-Explaining on Learning and Transfer of Critical Thinking Skills." *Frontiers in Education* 3: 100. <https://doi.org/10.3389/feduc.2018.00100>.
- Wang, Y., S. Gong, Y. Cao, and W. Fan. 2023. "The Power of Affective Pedagogical Agent and Self-Explanation in Computer-Based Learning." *Computers & Education* 195: 104723. <https://doi.org/10.1016/j.compedu.2022.104723>.

- Wehrmann, F., and R. Zender. 2025. "A Systematic Review of Cooperation in Multi-User Virtual Reality Learning Environments." *Journal of Computer Assisted Learning* 41, no. 5: e70112. <https://doi.org/10.1111/jcal.70112>.
- Wen, C. T., C. C. Liu, C. Y. Li, et al. 2024. "The Learning Analytics of Computational Scientific Modeling With Self-Explanation for Subgoals and Demonstration Scaffolding." *Computers & Education* 215: 105043. <https://doi.org/10.1016/j.compedu.2024.105043>.
- Wu, X., X. Chen, J. Zhao, and Y. Xie. 2023. "Influences of Design and Knowledge Type of Interactive Virtual Museums on Learning Outcomes: An Eye-Tracking Evidence-Based Study." *Education and Information Technologies* 29, no. 6: 7223–7258. <https://doi.org/10.1007/s10639-023-12061-4>.
- Zhang, Y., C. Liu, Y. Xing, Z. Pi, and J. Yang. 2024. "How Does Drawing Influence the Effectiveness of Oral Self-Explanation Versus Instructional Explanation in Video Learning?" *British Journal of Educational Technology* 55, no. 3: 1189–1208. <https://doi.org/10.1111/bjet.13423>.
- Zhou, Y., J. Chen, and M. Wang. 2022. "A Meta-Analytic Review on Incorporating Virtual and Augmented Reality in Museum Learning." *Educational Research Review* 36: 100454. <https://doi.org/10.1016/j.edurev.2022.100454>.